



PSOC™ Edge Trusted Firmware-M

Customer training workshop (CTW)

April 2026



Prerequisites

– Training prerequisites

- Introduction to PSOC™ Edge and ModusToolbox™
 - For an introduction to PSOC™ MCUs, including a getting started guide, visit <https://www.infineon.com/product-information/psocdeveloper>
 - For PSOC™ Edge trainings, visit [PSOC™ Edge E84 Training Collection](#)
 - Introduction to PSOC™ Edge Security: <https://github.com/Infineon/mtb-training-psoc-edge-security-intro>

– Development tools

- ModusToolbox™ v3.7
- Eclipse IDE for ModusToolbox™ v2026.1.0
- Edge Protect Security Suite v1.6
- ModusToolbox™ Programming Tools v1.7.0
- KIT_PSE84_EVAL_EPC2 BSP v1.0.0
- Terminal Emulator
- KIT_PSE84_EVAL

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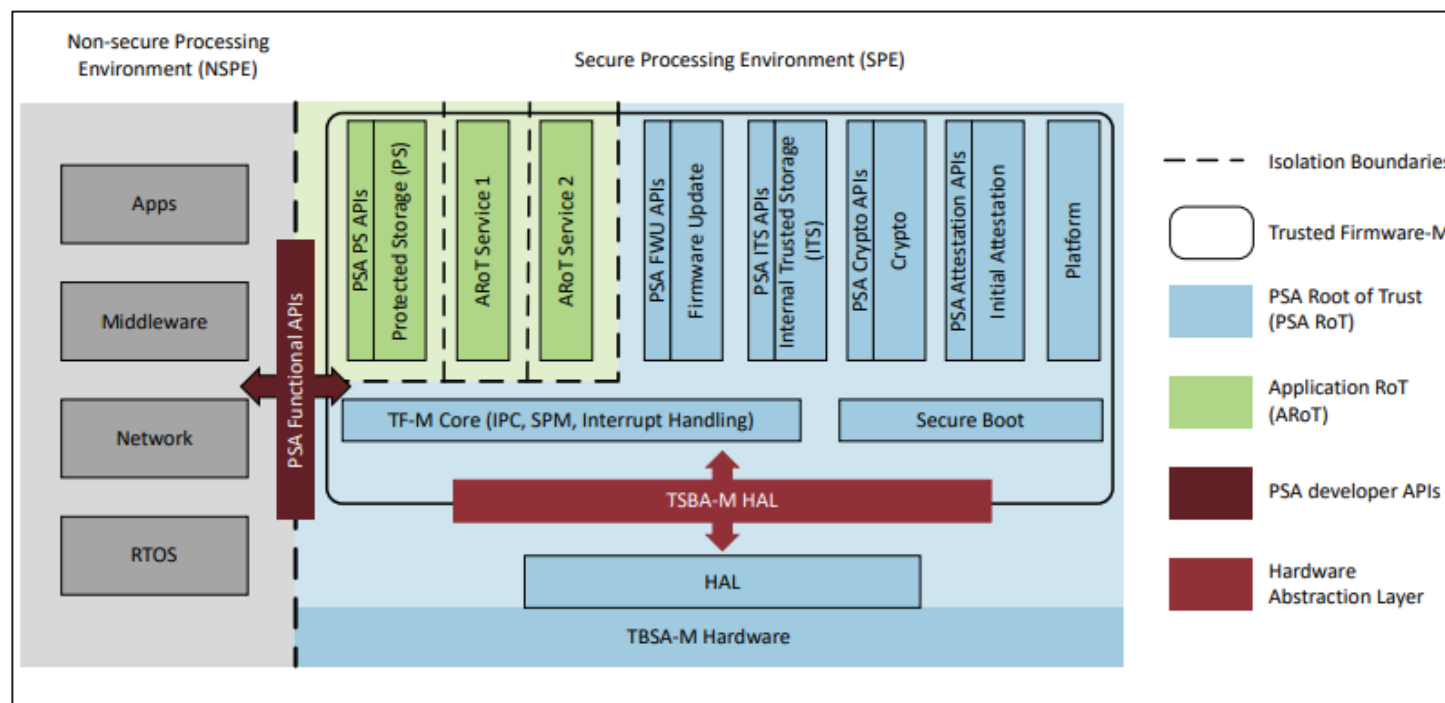
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Trusted Firmware-M (TF-M) overview

- Trusted Firmware-M (TF-M) provides a Secure processing environment (SPE) for Armv8-M, Armv8.1-M, and dual-core platforms
- It serves as the reference implementation of the Platform Security Architecture (PSA), aligning with PSA Certified guidelines to enable devices to achieve PSA Certification



- **Note:** For more information, visit: <https://trustedfirmware-m.readthedocs.io/en/latest/introduction/readme.html>

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TF-M architecture

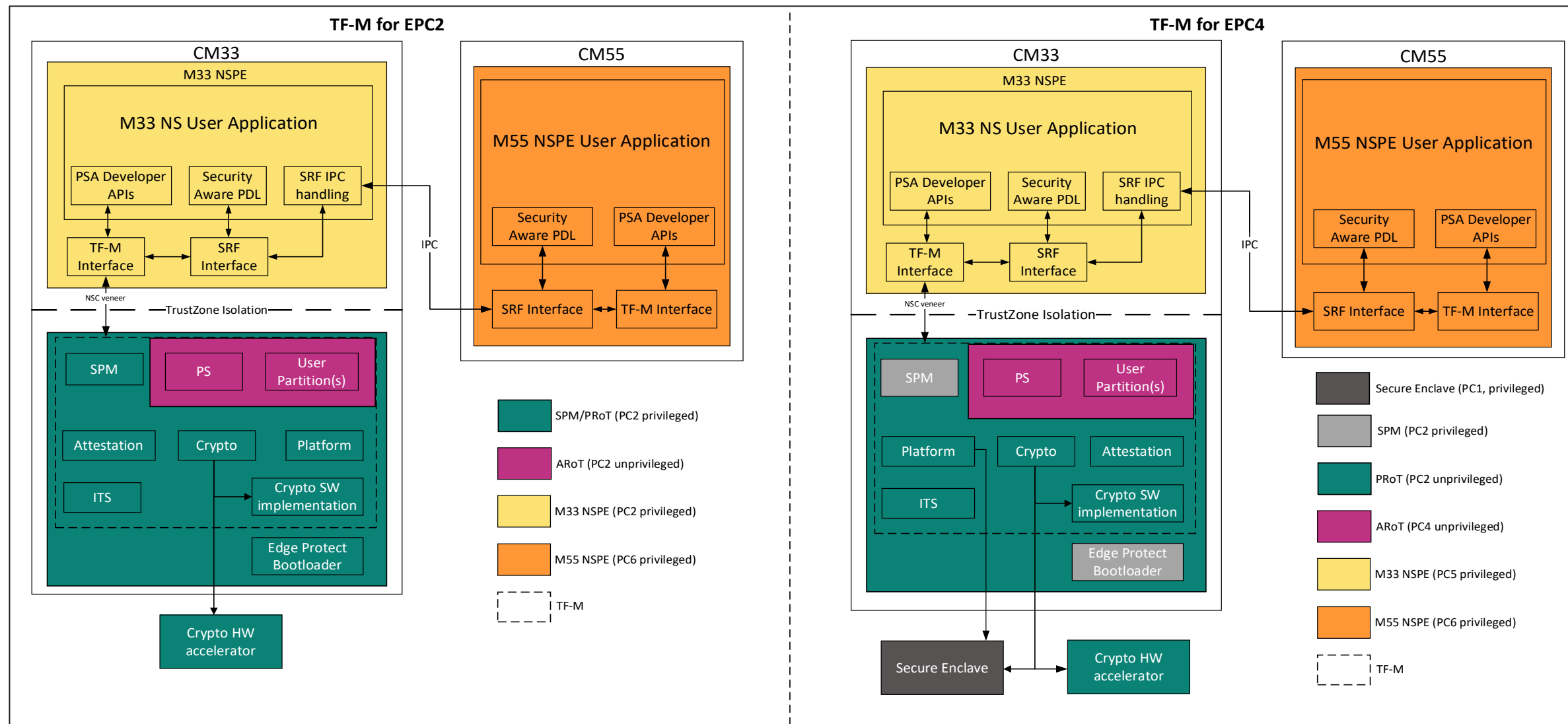


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Boot flow with TF-M

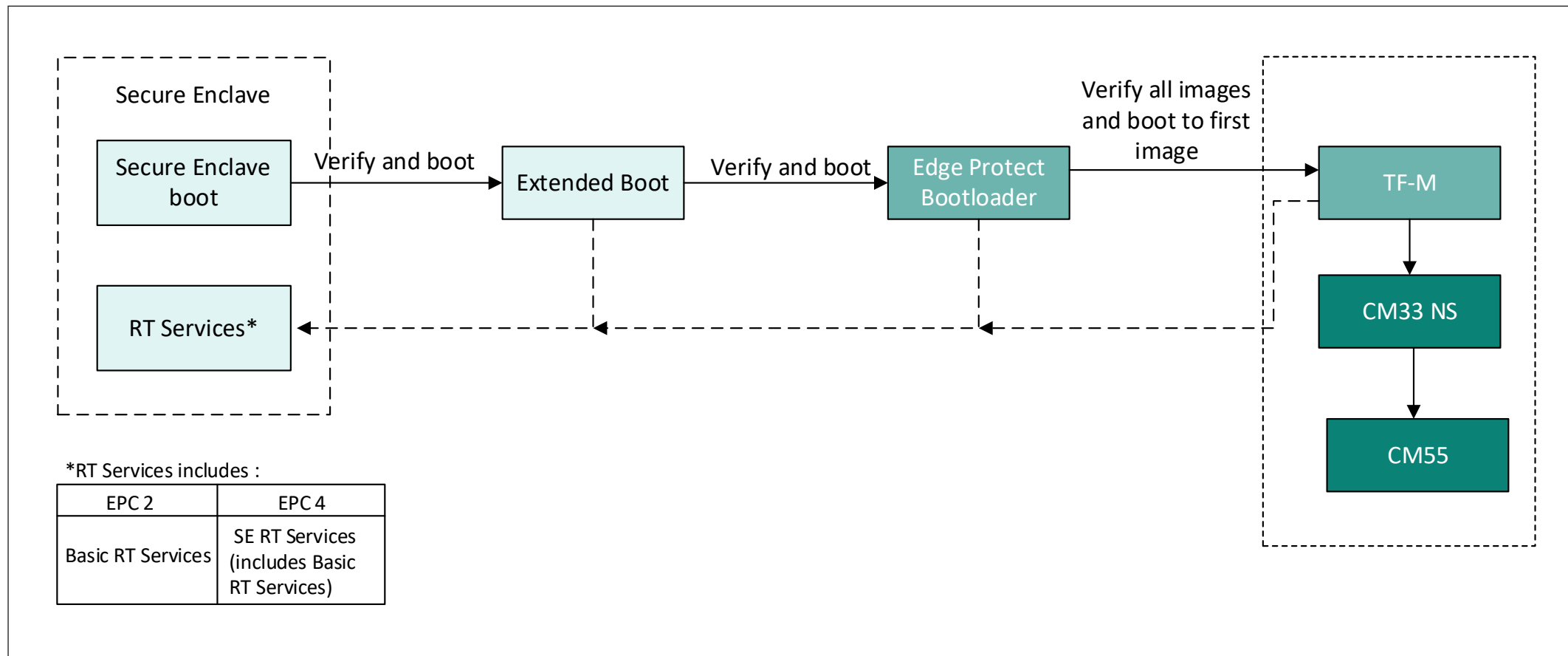


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Isolation levels

The PSA Firmware Framework-M (PSA FF-M) specification defines three isolation levels:

- **Level 1**: An isolation boundary exists between SPE and NSPE
- **Level 2**: Isolation boundary exists between SPE and NSPE, and between PSA RoT and App RoT
- **Level 3**: Isolation boundary exists between SPE and NSPE, between PSA RoT and App RoT, and among individual App RoT partitions

Device compliance

- **EPC 2**: PSA L2 compliant and supports isolation level 2 (L2) and isolation level 3 (L3), with L2 being the default isolation level
- **EPC 4**: Supports both L2 and L3, with L3 as default

EPC 2 and EPC 4 privilege levels

Privilege level of EPC2 software component

Software component	NS attribute	PC	Priv attribute
Edge Protect Bootloader	S	PC2	Priv
TF-M core (SPM)	S	PC2	Priv
PSA RoT	S	PC2	Priv
App RoT	S	PC2	Unpriv
M33 NSPE	NS	PC2	Priv
M55 NS application	NS	PC6	Priv

Privilege level of EPC4 software component

Software component	NS attribute	PC	Priv attribute
SE RT services	S	PC1	Priv
Edge Protect Bootloader	S	PC2	Priv
TF-M core (SPM)	S	PC2	Priv
PSA RoT	S	PC2	Unpriv
App RoT	S	PC4	Unpriv
M33 NSPE	NS	PC5	Priv
M55 NS application	NS	PC6	Priv

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TF-M profiles

	Small (not supported)	Medium	ARoT less (not supported)	Large
Isolation level	Level 1	Level 2	Level 1	Level 3
Crypto capabilities	• Symmetric cipher suite • SHA 256 hashing	• Symmetric and asymmetric cipher suite • SHA 256 hashing	• Symmetric and asymmetric cipher suite • SHA 256 hashing	• Larger symmetric and asymmetric cipher suite • SHA 256/384/512 hashing
Services/partitions	• ITS • Initial Attestation (symmetric) • Crypto	• ITS • Initial Attestation (asymmetric) • PS • Crypto	• ITS • Initial Attestation (asymmetric) • FWU • Crypto	• ITS • Initial Attestation (asymmetric) • PS • Crypto • FWU (not supported)

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TF-M services: Cryptography

- TF-M offers a **Cryptography** service implementing the PSA crypto API within a PSA RoT secure partition using the MbedTLS library
- The crypto service is accessible to applications in NSPE and other TF-M services in SPE via PSA Crypto APIs
- TF-M supports four build profiles, each with different functionality:
 - **Small** (not supported on PSOC™ Edge)
 - **Medium**
 - **ARoT less** (not supported on PSOC™ Edge)
 - **Large**
- By default, PSOC™ Edge EPC 2 and EPC 4 devices use the medium profile

TF-M services: Cryptography – EPC 2

- The PSOC™ Edge EPC 2 device implements cryptography using the following:
 - A crypto hardware accelerator for accelerated algorithms
 - Software implementation from the Mbed TLS library for other algorithms
- The choice between hardware and software implementation is determined at compile-time configuration
- By default, hardware accelerator support is enabled in TF-M for EPC 2 devices. This support can be enabled or disabled by editing the application [Makefile](#)
- For key storage
 - Persistent keys are stored in Internal Trusted Storage (ITS)
 - Volatile keys are stored in an SRAM region allocated to the Crypto partition

TF-M services: Cryptography – EPC 4

The crypto implementation for the EPC 4 device uses the following:

- Secured enclave (SE) for robust security
- Crypto hardware accelerator for accelerated algorithms (PSA L2 certifiable)
- Software implementation from the Mbed TLS library for other algorithms

Recommendation:

- Use Secure Enclave (SE) for crypto operations and keep the crypto hardware accelerator disabled. Enabling the crypto hardware accelerator prevents compliance with PSA L3/4

Key management:

- EPC 4 uses the key lifetime attribute to determine whether SE or M33 software handles crypto operations. When SE is used, both persistent and volatile keys are stored inside SE
- For SHA256 hashing, two options are available with EPC 4 device: SE or software implementation. Unlike other crypto operations, the SHA256 selection must be done at compile time. By default, SHA256 with SE is enabled

TF-M services: Secured storage

Internal trusted storage (ITS)

- ITS is a PSA RoT service designed for storing the most security-critical device data, such as cryptographic keys, in the internal RRAM storage
- In PSOC™ Edge, RRAM NVM is used to store the Internal trusted storage assets. By default, Internal trusted storage supports the storage of ten assets, with a maximum asset size of 512 bytes

Protected storage (PS)

- PS is an application RoT service that offers encrypted storage of assets in the internal RRAM
- PS uses AES-CCM-based AEAD encryption to protect the assets
- By default, PS supports the storage of ten assets, with a maximum asset size of 2 KB

TF-M services: Initial attestation

- The initial attestation service serves as a key mechanism to prove the genuineness of the device
- Upon request, TF-M can create an initial attestation token (IAT) that is compliant with the [IETF entity attestation token \(EAT\)](#) specification
- The attestation token contains a list of claims that serve as evidence to prove the identity of the device
- The following list of claims is provided in the attestation token:
 - boot seed
 - challenge
 - client_idimplementation_id
 - instance_id
 - profile_id
 - security_lifecycle
 - sw_components (for each image): measurement_description; measurement_value; signer_id; sw_component_type; sw_component_version

TF-M services: Initial attestation – EPC 2 / EPC 4

- The EPC 2 device includes software component measurements for the following images:
 - CM33 SPE (TF-M)
 - CM33 NSPE
 - CM55 NSPE

- The EPC 4 device includes software component measurements for the following images:
 - SE RRAM boot
 - SE RT services
 - Extended boot (L1 boot)
 - CM33 SPE (TF-M)
 - CM33 NSPE
 - CM55 NSPE

TF-M platform services

TF-M for PSOC™ Edge offers the following platform services.

- Non-volatile counters

TF-M offers up to three non-volatile counters of four bytes each to the NSPE.

- System reset

The registers required to reset the system are protected and cannot be accessed by the NSPE. The system reset service provides a way for the NSPE application to reset the platform.

- NSPE logging

The logging platform service can be used by the NSPE to log messages on the serial terminal using UART.

- Security-aware peripheral driver library (PDL)

Infineon's port of TF-M has a Secure Request framework (SRF) integrated as part of the platform partition. It provides a security-aware driver implementation of SMIF, RTC, SysClk, and SysPM modules.

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Default configurations for EPC 2/EPC 4

Configuration	EPC2	EPC4
Profile	Medium	Medium
Isolation level	Level 2	Level 3
Services/Partitions	• ITS • Initial attestation (asymmetric) • PS • Crypto • Platform	• ITS • Initial attestation (asymmetric with Secure Enclave) • PS • Crypto (with Secure Enclave) • Platform
NV counters location	RRAM	Secure Enclave

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Application structure with TF-M

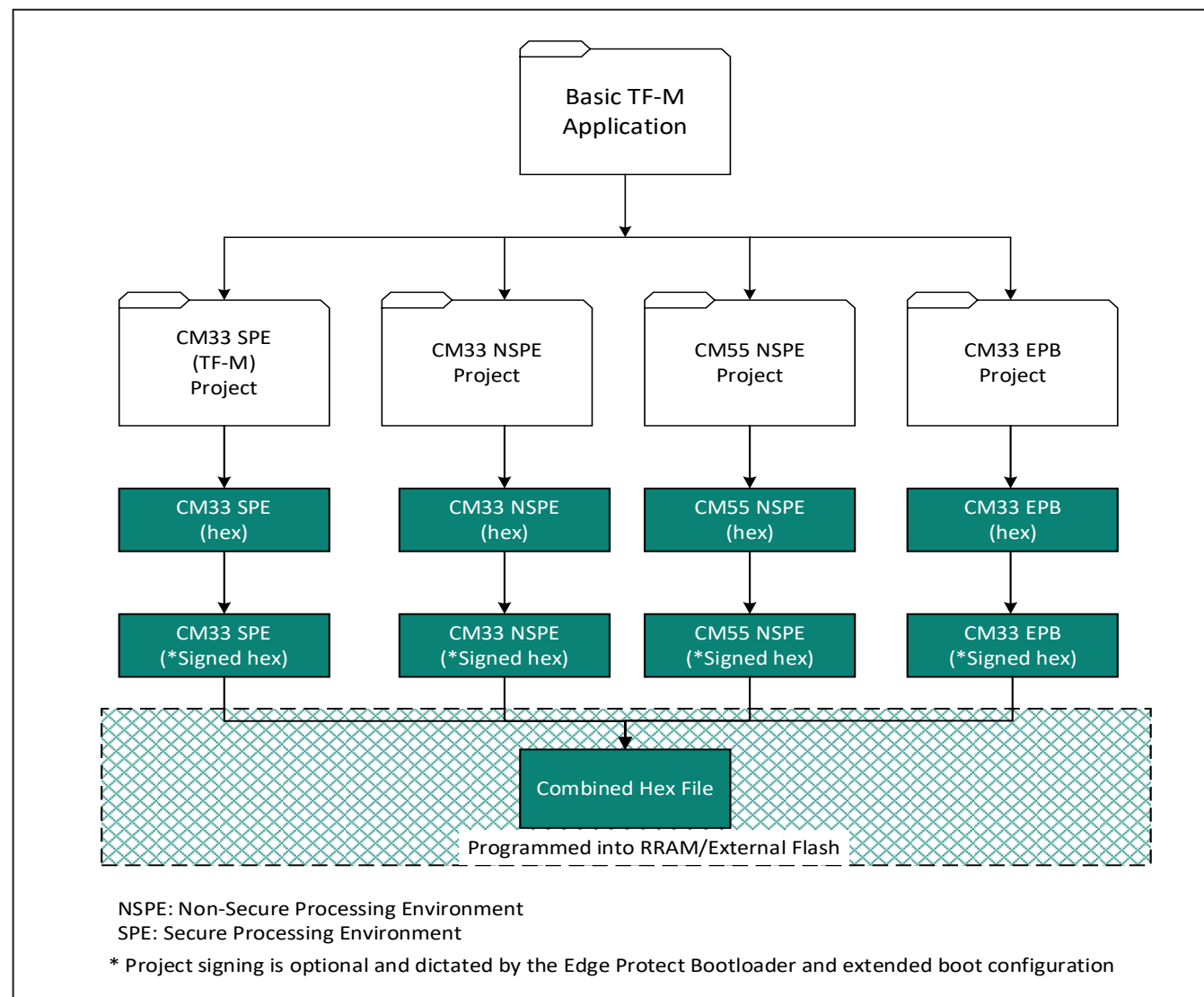


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Lab 1: TF-M Crypto

What will you learn?

- How to boot to a TF-M integrated application
- How to calculate the hash, generate keys, sign, verify, encrypt and decrypt a message using the PSA crypto APIs.

Lab 2: Adding new application RoT partition

What will you learn?

- How to add a new partition using Edge Protect Configurator
- How to generate files for the partition
- How to customize the services in the partition and invoke the service API

Revision history

Document revision	Date	Description of change
**	2026-01-14	Initial release.
*A	2026-04-28	Updated speaker notes. Updated version of tools. Added slides for labs. Updated images to align with brand guidelines.

